

Image restoration method based on fractional variable order differential

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Abstract In this paper, we study image restoration problem by using fractional variable order differential technique. Our idea is to make use of fractional order differential diffusion equation of evolution procedure into the image restoration problem. An image often exists different low-frequency and high-frequency components, such as flat, texture and edge, etc. Because fractional order differential can enhance the high-frequency components of a signal, meanwhile, nonlinearly preserve the low-frequency components of the signal, we can adapt suitable fractional differential orders to restore their components. In particular, different differential orders can be used in an image at the same time. In order to obtain the restored image automatically, we choose the fractional differential orders by the value of gradient modulus of the image and use the discrete Fourier transform to implement the numerical algorithm. We also provide an iterative scheme in the frequency domain. Experimental results are reported to demonstrate that the visual effects, the combined image similarity index and the peak signal to noise ratio of restored images by using the proposed method are very good, and are competitive with the other testing methods.

Keywords Image restoration · Fractional · Variable order differential · Total variation · Combined image similarity index

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1 Introduction

The quality of the images may be degradation in the process of image formation, transmission and storage (Guillemot and Meur 2014; Chen 2015; Zhang et al. 2014a; Yan and Lu 2015; Bini and Bhat 2014). For example, it can make the quality of images degradation such as the object's movement, the defects of imaging systems, the inherent noise of recording equipment and external disturbance and so on. Assume that imaging system is linear translation invariant system, then the degradation of the image can be described by the mathematical model as the following (Guillemot and Meur 2014; Huang 2013; Wang et al. 2016b):

$$u_0 = h_d * f + n \quad (1)$$

where $f(x, y)$ denotes the ideal image, and $h_d(x, y)$ denotes the point-spread function (PSF) of the imaging system. The ideal PSF of imaging system is $\delta()$ function. Due to various reasons, the PSF of the actual imaging system is deviated from the $\delta()$ function. However, in principle, we can through theoretical analysis and experimental measurement for imaging system to estimate the PSF of the system (Wang et al. 2016a; Wiener 1942). For example, Defocus can form blur, and the corresponding PSF can approximate to Gaussian function:

$$h_d(x, y) = \exp \{-(x^2 + y^2)/(2\delta^2)\}. \quad (2)$$

The influence of atmospheric turbulence can be modeled as:

$$H_d(u, v) = \exp \{-k(u^2 + v^2)^{5/6}\} \quad (3)$$

where $H_d(u, v)$ denotes the Fourier transform of $h_d(x, y)$, and it is the transfer function of imaging system. We can find that the index in Eq. (3) is $5/6 \approx 1$, so the expression is only slightly deviated from Gaussian function. We look at the motion blur. When objects are constant-velocity linear motion, there is

$$H_d(u, v) = T \frac{\sin [\pi (au + bv)]}{\pi (au + bv)} \exp \{-j\pi (au + bv)\} \quad (4)$$

where T denotes shutter time, a and b denote x and y component of the object movement speed, respectively. If readers want to learn more about PSF of estimate system, please refer to the Ref. (Wiener 1942). We will assume that the degraded function h_d is known in the following. We use n to denote the additive noise of the image (Wiener 1942; Castlemen 1996), and assume it is Gauss white noise model with zero mean and σ variance.

It is the image restoration problem that we try to restore the original appearance of the ideal image f in Eq. (1) by the degraded images u_0 which has been recorded and the prior knowledge of the degraded system. It is one of the classic problem in image processing.

The purpose of this paper is to design a new model of image restoration so that the restored image preserve the edge, texture and details of the ideal image as much as possible. The rest of this paper is organized as follows. Section 2 introduces the traditional method of image restoration and the TV based image restoration model. Combined fractional variable order differential, we put forward a new image restoration method based on fractional variational order differential equation and analyze it in Sect. 3. In Sect. 4, the results and analysis of our experiment are given. Finally, we present the conclusions and prospects of our research in Sect. 5.

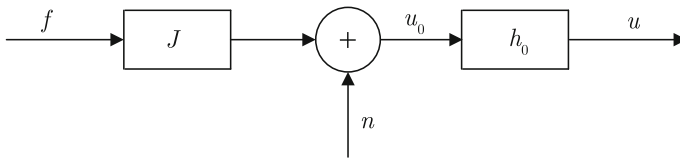


Fig. 1 Image restoration deconvolution filter method

2 Preliminaries

This section consists of two parts. One is the traditional method of image restoration, and another is the TV based image restoration model.

2.1 The traditional method of image restoration

Traditional image restoration methods are mostly based on linear system as shown in Fig. 1,

In the case of ignoring faint noise in image, according to the convolution theorem, we can obtain the following formula by the degraded model formula (1):

$$U_0(u, v) = F(u, v)H_d(u, v). \quad (5)$$

Thereby, we have

$$F(u, v) = U_0(u, v) \frac{1}{H_d(u, v)} \quad (6)$$

That is to say, we can obtain the restored image $f(x, y)$ as long as the degraded image $u_0(x, y)$ via a linear filter whose transfer function is

$$H_0(u, v) = 1/H_d(u, v). \quad (7)$$

We usually call this approach as 'deconvolution' or 'inverse filtering'. This approach looks simple and easy to do, but in fact, it has not use value. Main reason is that the inverse filter defined by the above formula may not exist (because the H_d in some high frequency may be zero). And even if such an inverse filter is exist, the low-pass property of H_d can lead to the result that the value of $|H_0(u, v)|$ is very large in the high frequency region. It would amplify the weak noise which could be ignored.

The starting point of Wiener deconvolution filter theory (Wiener 1942) is to make the mean square error of the output image u and the ideal image f reach to minimum, namely, minimize the functional:

$$E\{h\} = E\{(f - u)^2\} = E\{(f - h * u_0)^2\} = E\{(f - h * (J * f + n))^2\} \quad (8)$$

where $E\{\}$ denotes mathematical expectation. If both f and n are stationary random process, and the mean of one of them is zero, then we can obtain the solution of the above formula as (Richtmyer and Morton 1967)

$$h_0 * (h_d * \bar{h}_d * R_f + R_n) = \bar{h}_d * R_f \quad (9)$$

where R_f and R_n denote the ideal image f and the autocorrelation function of noise n , respectively. Using FT, we can obtain

$$\begin{aligned} H_0(u, v) &= \frac{\bar{H}_d(u, v) P_f(u, v)}{|H_d(u, v)|^2 P_f(u, v) + N(u, v)} \\ &= \frac{1}{H_d(u, v)} \cdot \frac{|H_d(u, v)|^2}{|H_d(u, v)|^2 + N(u, v)/P_f(u, v)} \end{aligned} \quad (10)$$

where $P_f(u, v)$ and $N(u, v)$ denote power spectrums of f and n , respectively. For the white noise, N is constant. But usually, we don't have the prior knowledge of $P_f(u, v)$. Thus, in the specific design, we must make further assumptions. By the Eq. (10), it is not hard to find that we can see the Wiener deconvolution filter $H_0(u, v)$ as a cascade of the inverse filter and Wiener filter, where the transfer function of Wiener filter is

$$Q_2(u, v) = \frac{|H_d(u, v)|^2}{|H_d(u, v)|^2 + N(u, v)/P_f(u, v)}. \quad (11)$$

Another important classic method of image restoration is constrained least squares method. Its starting point is that, in the following condition, output image is smooth as much as possible

$$\iint_{\Omega} |h_d \cdot u - u_0|^2 dx dy = \sigma^2 = \text{const} \quad (12)$$

If we use the $\iint_{\Omega} |\nabla u|^2 dx dy$ as the measure of image smoothness, it will constitute a variational problem which has the following functional:

$$E(u) = \iint_{\Omega} |\nabla u|^2 d\Omega + \lambda \iint_{\Omega} |h_d \cdot u - u_0|^2 d\Omega \quad (13)$$

where λ denotes Lagrange multiplier. It is not difficult to obtain the Euler–Lagrange equation by minimizing Eq. (13) as follows:

$$\mu \Delta u + \bar{h}_d(h_d \cdot u - u_0) = 0 \quad (14)$$

where $\mu = 1/\lambda$. Operating Fourier transform both sides of the above formula, we can obtain

$$\mu(\omega_x^2 + \omega_y^2)\hat{u} + |H_d|^2\hat{u} - \hat{J}\hat{u}_0 = 0 \Rightarrow \hat{u} = \frac{\hat{J}\hat{u}_0}{\mu(\omega_x^2 + \omega_y^2) + |H_d|^2}.$$

Obviously, there is the corresponding linear filter for the constrained least squares restoration method as follows

$$H_0(u, v) = \frac{\bar{J}(u, v)}{|H_d(u, v)|^2 + \mu(u^2 + v^2)} = \frac{1}{H_d(u, v)} \frac{|H_d(u, v)|^2}{|H_d(u, v)|^2 + \mu(u^2 + v^2)} \quad (15)$$

It can also be viewed as the cascading of inverse filter and other filter. Compared with Wiener deconvolution filter, the difference is that it can still complete the design of filter in the case of no a priori knowledge about ideal image power spectrum. A lot of experiments show that if the selecting of the parameter μ is appropriate, the the performance of constrained least squares restoration method will be slightly better than the Wiener deconvolution filter method (Gonzalez and Woods 2003).

Other traditional methods of image restoration are mostly variation of the above methods. Here, we do no longer introduce them one by one. In generally, the effect of the traditional methods of image restoration is not satisfactory. Especially in the case of strong noise, the

output image u either appears a ‘ringing’ phenomenon or appear obviously residual noise and fuzzy phenomenon.

2.2 TV based image restoration model

2.2.1 Variational bounded function space and total variation norm

Variational bounded function space is definition as (Wang et al. 2008)

$$BV(\Omega) := \left\{ u, \int_{\Omega} |Du| d\Omega < \infty \right\} \quad (16)$$

where Du denotes distributional derivative of u (Aubert and Kornprobst 2002). For one dimension, the total variation of the function of bounded variation is defined as follows:

$$TV(u) := \int_{\Omega} |Du| dx \quad (17)$$

It can also be abbreviated as

$$TV(u) = \int_{\Omega} |u_x| dx \quad (18)$$

where, u_x is interpreted as finite difference quotient

$$\frac{u(x+h) - u(x)}{h}$$

We generalize directly the above definition to two dimensions or higher dimensions, there is

$$TV(u) = \int_{\Omega} |\nabla u| d\Omega \quad (19)$$

According to the defined formula of one dimensional total variation Eq. (18), we can obtain an important property of $TV(u)$, namely, if $u \in BV([a, b])$, and u is monotonous in $[a, b]$ and meets $u(a) = \alpha$, $u(b) = \beta$, no matter what the detailed form of function u , there will always be

$$TV(u) = |\alpha - \beta|. \quad (20)$$

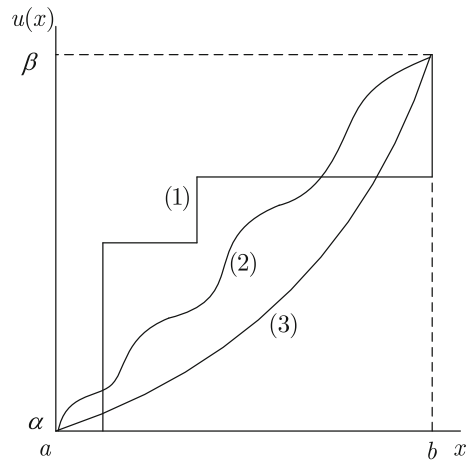
According to Eq. (18), Assume that u are derivable in the boundary point a and b , the above results can be obtained. Figure 2 show three function curves whose smoothness difference are large very much. However, according to the formula (20), their total variation are the same. This means that if we use the total variational $TV(u)$ to measure the smoothness of function u , the three curves are the same smoothness. Therefore, even if u exist ‘jump’, we can also reduce the total variation $TV(u)$. Thus, it is possible that the edge of the image(jump) is preserved in the process of minimizing total variation. But if we use the $\int |\nabla u|^2$ as a measure of the image “smoothness”, for the three functions in Fig 2, obviously, there is the following inequality relationship

$$\int |\nabla u_1|^2 > \int |\nabla u_2|^2 > \int |\nabla u_3|^2$$

From this point of view, curve (3) is the most smooth, curve (1) is the most non-smooth, curve (2) is in between of them. As a result, when we minimize $\int |\nabla u|^2$, the larger “jumping” will be smooth firstly.

To sum up, if the image is classified as variation bounded function space, namely, $u \in BV(\Omega)$, and fractional order norm total variation $\int |\nabla u_1|^\alpha$ serves as the measure of the image “smoothness”, it will be a very appropriate mathematical model in image processing.

Fig. 2 The property of total variation



2.2.2 TV based image restoration model

Total variation(TV) based image restoration models were first introduced by Rudin, Osher, and Fatemi (ROF) in their pioneering work (Rudin et al. 1992) on edge preserving image denoising. It is one of the earliest and best known examples of PDE based edge preserving denoising. It was designed with the explicit goal of preserving sharp discontinuities (edges) in images while removing noise and other unwanted fine scale detail. TV model is one of the simplest variational models having this most desirable property. The revolutionary aspect of this model is its regularization term that allows for discontinuities but at the same time disfavors oscillations. It was originally formulated in Ref. (Rudin et al. 1992) for grayscale imagery in the following form:

$$\inf_{\int_{\Omega} (u-f)^2 dx = \sigma^2} \int_{\Omega} |\nabla u|. \quad (21)$$

Where, Ω denotes the image domain (for instance, the computer screen), and is often a rectangle. The function $f(x) : \Omega \rightarrow \mathbb{R}$ represents the given observed image, which is assumed to be corrupted by Gaussian noise of variance σ^2 . The constraint of the optimization forces the minimization to take place over images that are consistent with this known noise level. The objective functional itself is called the total variation of the function $u(x)$; for smooth images it is equivalent to the L^1 norm of the derivative, and hence is some measure of the amount of oscillation found in the function $u(s)$. Optimization problem (21) is equivalent to the following unconstrained optimization, which was also first introduced in Ref. (Rudin et al. 1992):

$$\inf_{u \in L^2(\Omega)} \int_{\Omega} |\nabla u| + \lambda \int_{\Omega} (u-f)^2 dx. \quad (22)$$

Where, $\lambda \geq 0$ is a Lagrange multiplier. The equivalence of problems (21) and (22) has been established in (Chambolle and Lions 1997).

In the above TV image restoration model, there are two shortcomings: First, its Euler-Lagrangian equation is a partial differential equation with ill-conditions; Second, It is not completely conform to the morphological principle of image processing. With the evolution of images, the changes not only depend on the level set (characterized by ∇u), but also depends on the grey value u at the same time. Direct performance of this defect is that there is obvious 'piecewise constant' effect, or called 'ladder' effect in its steady-state solution. In

order to solve the problem, we proposed image restoration model based on fractional variable order differential model in the next section.

3 Methodology

In this section, a new image restoration model based on fractional variable order differential is proposed and analyzed.

3.1 Image restoration model based on fractional variable order differential

We first introduce the concept of fractional variable order differential (Zhang et al. 2014b).

Definition 1 Supposing $\vec{d} = (d_1, d_2, \dots, d_n)^T$ is a $n \times 1$ vector, $\mathbf{A} = (a_{ij})_{n \times m}$ is a $n \times m$ matrix, $\vec{f}(t) = (f_1(t), f_2(t), \dots, f_n(t))^T$ is a $n \times 1$ function vector and $\mathbf{u}(x, y) = (u_{ij}(x, y))_{n \times m}$ is a $n \times m$ function matrix, D_α is α -th ($\alpha \in R^+$) fractional derivative operator. Then we define:

$$\vec{d}^{\mathbf{A}} := \begin{pmatrix} d_1^{a_{11}} & d_1^{a_{12}} & \dots & d_1^{a_{1m}} \\ d_2^{a_{21}} & d_2^{a_{22}} & \dots & d_2^{a_{2m}} \\ \vdots & \vdots & \ddots & \vdots \\ d_n^{a_{n1}} & d_n^{a_{n2}} & \dots & d_n^{a_{nm}} \end{pmatrix},$$

$$\vec{f}^{(\mathbf{A})}(t) := \begin{pmatrix} f_1^{(a_{11})}(t) & f_1^{(a_{12})}(t) & \dots & f_1^{(a_{1m})}(t) \\ f_2^{(a_{21})}(t) & f_2^{(a_{22})}(t) & \dots & f_2^{(a_{2m})}(t) \\ \vdots & \vdots & \ddots & \vdots \\ f_n^{(a_{n1})}(t) & f_n^{(a_{n2})}(t) & \dots & f_n^{(a_{nm})}(t) \end{pmatrix},$$

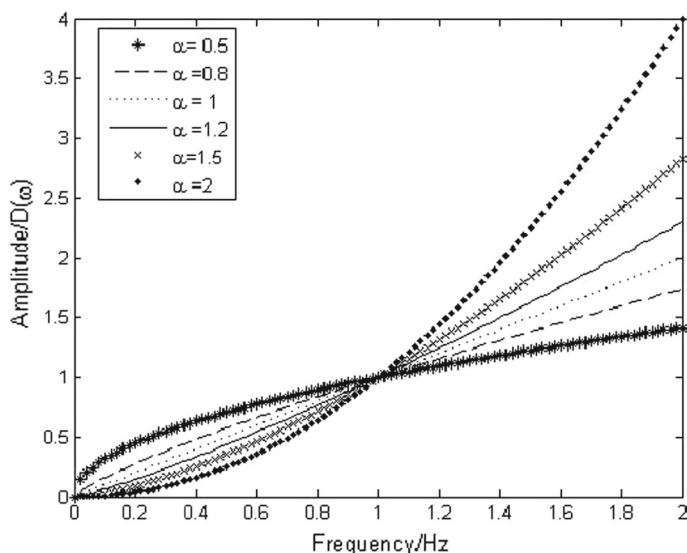


Fig. 3 The amplitude-frequency curve of fractional order differential

$$D_A := \begin{pmatrix} D_{a_{11}} & D_{a_{12}} & \cdots & D_{a_{1m}} \\ D_{a_{21}} & D_{a_{22}} & \cdots & D_{a_{2m}} \\ \vdots & \vdots & \ddots & \vdots \\ D_{a_{n1}} & D_{a_{n2}} & \cdots & D_{a_{nm}} \end{pmatrix},$$

$$D_A \mathbf{u}(x, y) := \begin{pmatrix} D_{a_{11}} u_{11}(x, y) & D_{a_{12}} u_{12}(x, y) & \cdots & D_{a_{1m}} u_{1m}(x, y) \\ D_{a_{21}} u_{21}(x, y) & D_{a_{22}} u_{22}(x, y) & \cdots & D_{a_{2m}} u_{2m}(x, y) \\ \vdots & \vdots & \ddots & \vdots \\ D_{a_{n1}} u_{n1}(x, y) & D_{a_{n2}} u_{n2}(x, y) & \cdots & D_{a_{nm}} u_{nm}(x, y) \end{pmatrix},$$

we call D_A as fractional variable order differential operator.

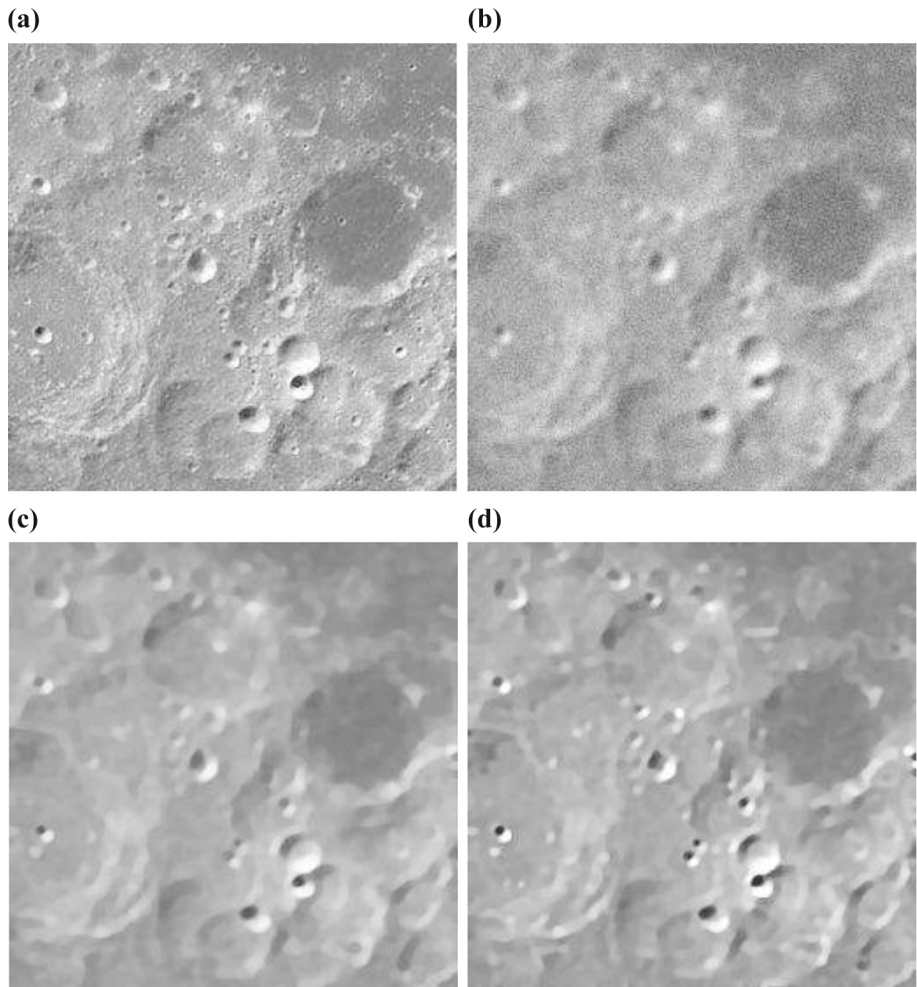


Fig. 4 Restoration results of comparison of TV model and our proposed model on Changeone. **a** Original image; **b** damaged image with Gaussian kernel blur and Gaussian Noise of $\sigma = 10$ (PSNR=24.1429, CISI=0.5211); **c** TV model (PSNR=25.9132, CISI=0.6770); **d** our proposed model (PSNR=26.2389, CISI=0.7032)

We consider the following model:

$$\operatorname{argmin}\{E(u)\}$$

where $E(u)$ is the following function defined in the space of continuous images over a support of Ω . This model is associated with the following energy function:

$$E(u) = \int_{\Omega} \rho(|D_A u|) d\Omega + \frac{\lambda}{2} \int_{\Omega} (h * u - u_0)^2 d\Omega, \quad (23)$$

where Ω denotes the image domain, and is usually a rectangle; $\lambda \geq 0$ is a Lagrange multiplier; The function $\rho(\cdot) \geq 0$ is an increasing function and meets the condition $\rho(0) = 0$; $A = \alpha(|\nabla u|)$ and $\alpha(\cdot)$ is an increasing function and satisfies the condition



Fig. 5 Restoration results of comparison of TV model and our proposed model on Lena. **a** Original image; **b** damaged image with Gaussian kernel blur and Gaussian Noise of $\sigma = 10$ (PSNR=24.1866, CISI=0.5301); **c** TV model (PSNR=27.0604, CISI=0.6932); **d** our proposed model (PSNR=27.3116, CISI=0.7161)

$$\alpha(x) \rightarrow \begin{cases} 1, & x \rightarrow 0 \\ 2, & x \rightarrow \infty \end{cases},$$

D_A denotes the fractional variable order differential operator defined by $D_A u = (D_{A_x} u, D_{A_y} u)$ and $|D_A u| = \sqrt{D_{A_x}^2 + D_{A_y}^2}$; h denotes Gaussian blur kernel (Wiener 1942). We can formally compute the Euler–Lagrange equation for this minimization problem as follows. By taking any test function $\eta \in C^\infty(\Omega)$ and assuming

$$\Phi(a) = \int_{\Omega} \rho(|D_A u + a D_A \eta|) dx dy + \frac{\lambda}{2} \int_{\Omega} (h * (u + a \eta) - u_0)^2 dx dy, \quad (24)$$

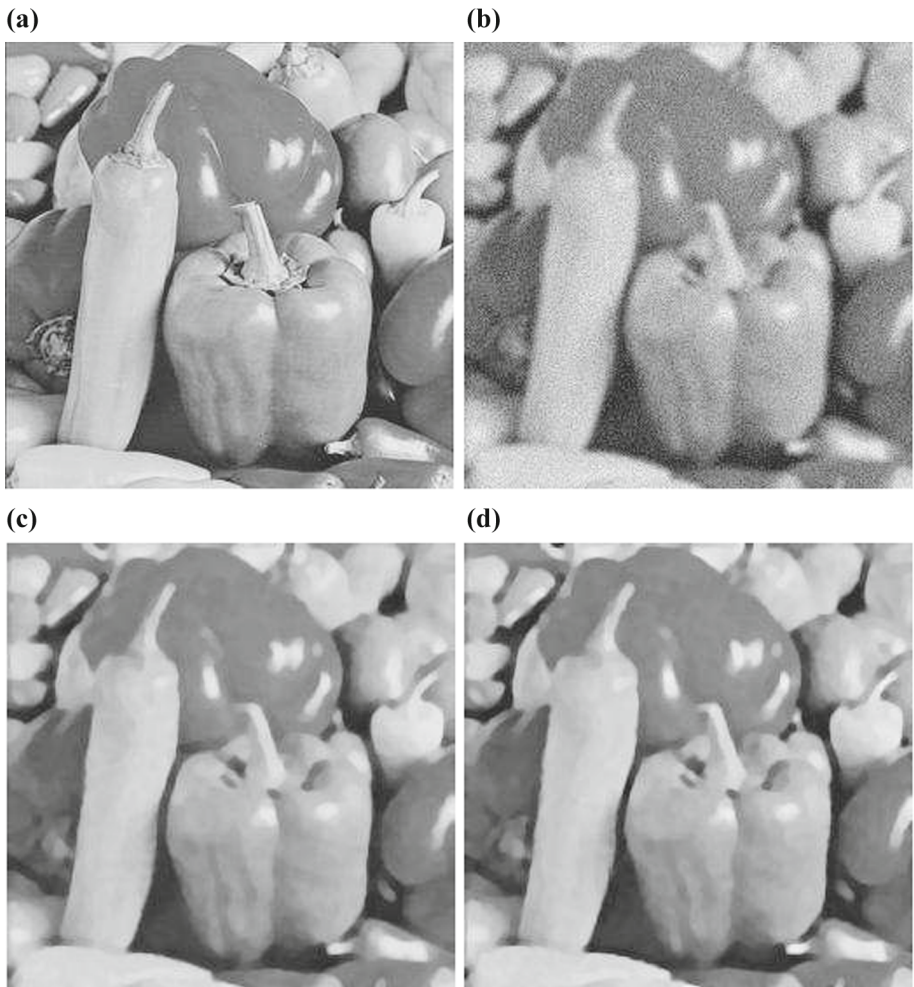


Fig. 6 Restoration results of comparison of TV model and our proposed model on Peppers. **a** Original image; **b** damaged image with Gaussian kernel blur and Gaussian Noise of $\sigma = 10$ (PSNR=24.2929,CISI=0.5331); **c** TV model (PSNR=27.6985,CISI=0.7163); **d** Our proposed model (PSNR=28.1243,CISI=0.7771)

we can obtain

$$\begin{aligned}
 \Phi'(0) &= \frac{d}{da} \left[\int_{\Omega} \rho(|D_{\mathbf{A}}u + aD_{\mathbf{A}}\eta|) dx dy + \frac{\lambda}{2} \int_{\Omega} (h * (u + a\eta) - u_0)^2 dx dy \right]_{a=0} \\
 &= \int_{\Omega} [\rho'(|D_{\mathbf{A}}u|) \frac{D_{\mathbf{A}_x}u}{|D_{\mathbf{A}}u|} D_{\mathbf{A}_x}\eta + \rho'(|D_{\mathbf{A}}u|) \frac{D_{\mathbf{A}_y}u}{|D_{\mathbf{A}}u|} D_{\mathbf{A}_y}\eta] dx dy \\
 &\quad + \lambda \int_{\Omega} (h * u - u_0)(h * \eta) dx dy \\
 &= \int_{\Omega} [c(|D_{\mathbf{A}}u|^2) D_{\mathbf{A}_x}u] D_{\mathbf{A}_x}\eta dx dy + \int_{\Omega} [c(|D_{\mathbf{A}}u|^2) D_{\mathbf{A}_y}u] D_{\mathbf{A}_y}\eta dx dy \\
 &\quad + \lambda \int_{\Omega} (h * u - u_0)(h * \eta) dx dy
 \end{aligned}$$

(a)



(b)



(c)



(d)



Fig. 7 Restoration results of comparison of TV model and our proposed model on Lady. **a** Original image; **b** damaged image with Gaussian kernel blur and Gaussian Noise of $\sigma = 10$ (PSNR=25.6635, CISI=0.6760); **c** TV model (PSNR=30.4625, CISI=0.8398); **d** our proposed model (PSNR=30.6072, CISI=0.8582)

$$= \int_{\Omega} \{D_{\mathbf{A}x}^*[c(|D_{\mathbf{A}}u|^2)D_{\mathbf{A}x}u] + D_{\mathbf{A}y}^*[c(|D_{\mathbf{A}}u|^2)D_{\mathbf{A}y}u] \\ + \lambda h * (h * u - u_0)\} \eta dx dy$$

for all $\eta \in C^\infty(\Omega)$, where $D_{\mathbf{A}x}^*$ is the adjoint of $D_{\mathbf{A}x}$ and $D_{\mathbf{A}y}^*$ is the adjoint of $D_{\mathbf{A}y}$ respectively; the coefficient $c(s) = \frac{\rho'(\sqrt{s})}{\sqrt{s}}$. Thus, the Euler–Lagrange equation is

$$D_{\mathbf{A}x}^*[c(|D_{\mathbf{A}}u|^2)D_{\mathbf{A}x}u] + D_{\mathbf{A}y}^*[c(|D_{\mathbf{A}}u|^2)D_{\mathbf{A}y}u] + \lambda h * (h * u - u_0) = 0. \quad (25)$$

The Euler–Lagrange equation may be solved through the following gradient descent procedure:

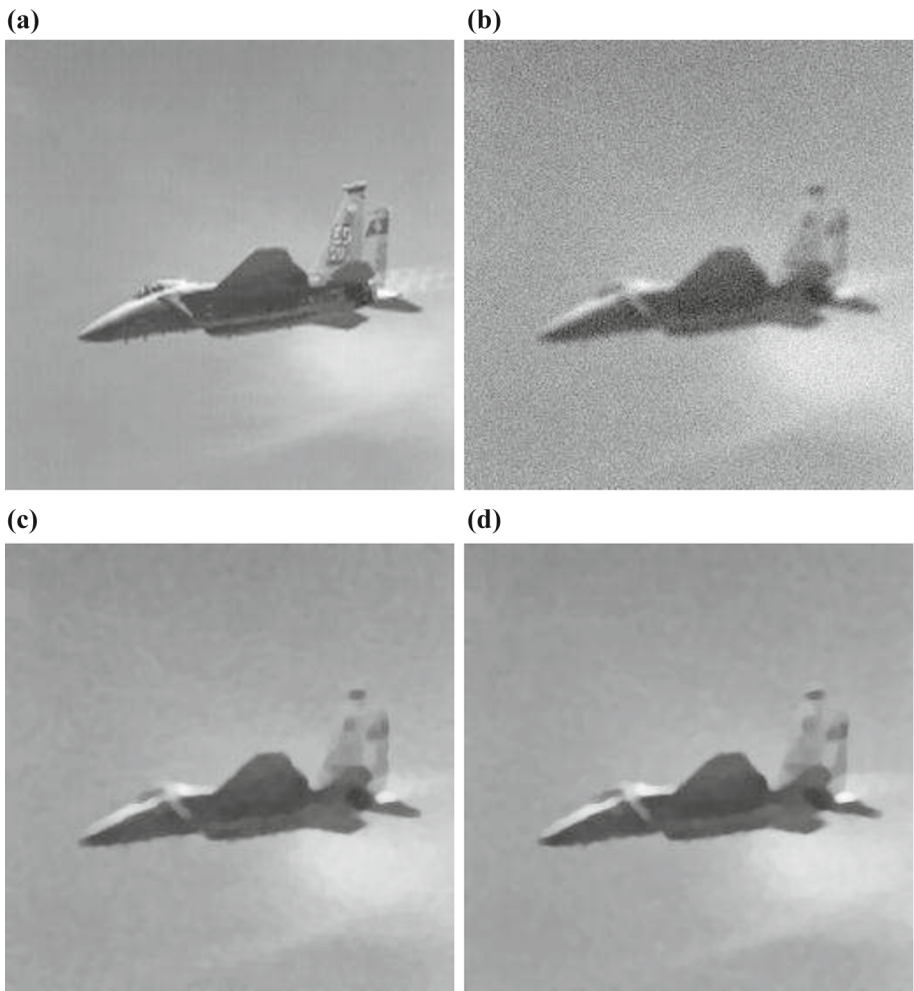


Fig. 8 Restoration results of comparison of TV model and our proposed model on Fighter. **a** Original image; **b** damaged image with Gaussian kernel blur and Gaussian Noise of $\sigma = 10$ (PSNR=27.2985,CISI=0.7407); **c** TV model (PSNR=35.1076,CISI=0.9359); **d** our proposed model (PSNR=35.4402,CISI=0.9467)

$$\begin{aligned} \frac{\partial u}{\partial t} = & -D_{\mathbf{A}x}^*[c(|D_{\mathbf{A}}u|^2)D_{\mathbf{A}x}u] - D_{\mathbf{A}y}^*[c(|D_{\mathbf{A}}u|^2)D_{\mathbf{A}y}u] \\ & -\lambda h * (h * u - u_0) \end{aligned} \quad (26)$$

The solution of the Euler–Lagrange equation is the restored image.

3.2 Analysis of the new model

In (Zhang et al. 2014b), we can know that the α th-order($\alpha \in R^+$) derivative of a function $f(t) \in L^2(R)$ in the frequency domain is

$$(D_\alpha f)(\omega) = (j\omega)^\alpha \hat{f}(\omega) = \hat{d}_\alpha(\omega) \hat{f}(\omega), \quad (27)$$

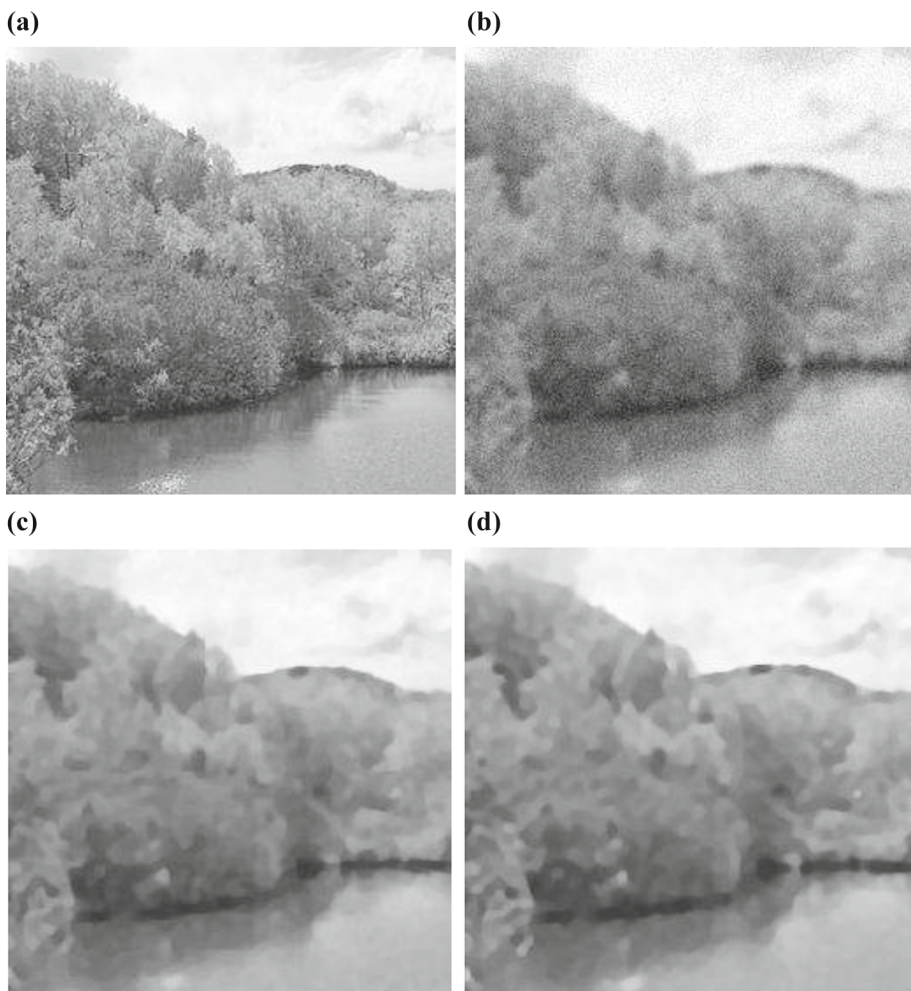


Fig. 9 Restoration results of comparison of TV model and our proposed model on Autumn. **a** Original image; **b** damaged image with Gaussian kernel blur and Gaussian Noise of $\sigma = 10$ (PSNR=24.2184,CISI=0.5321); **c** TV model (PSNR=26.0972,CISI=0.7001); **d** Our proposed model (PSNR=26.1313,CISI=0.7122)

where $\hat{f}(\omega)$ is the Fourier transform of $f(t)$, $\hat{d}_\alpha(\omega) = (j\omega)^\alpha$ is called as α th-order differential multiplier-function. Its complex exponential form is

$$\begin{cases} \hat{d}_\alpha(\omega) = \hat{a}_\alpha(\omega) \exp(j\theta_\alpha(\omega)) \\ \hat{a}_\alpha(\omega) = |\omega|^\alpha, \theta_\alpha(\omega) = \frac{\alpha\pi}{2} \operatorname{sgn}(\omega). \end{cases} \quad (28)$$

From Eq. (28), we can obtain the amplitude-frequency curves of fractional order differential, as shown in Fig. 3. From it, we find that the fractional order differential can enhance the high-frequency components of a signal, meanwhile, nonlinearly preserves the low-frequency components of the signal (Li and Xie 2016; Liu 2010; Che et al. 2013; Pu et al. 2010).

According to the feature of the fractional order differential for the signal, using different differential orders to process an image will produce different effects. Therefore, the proposed

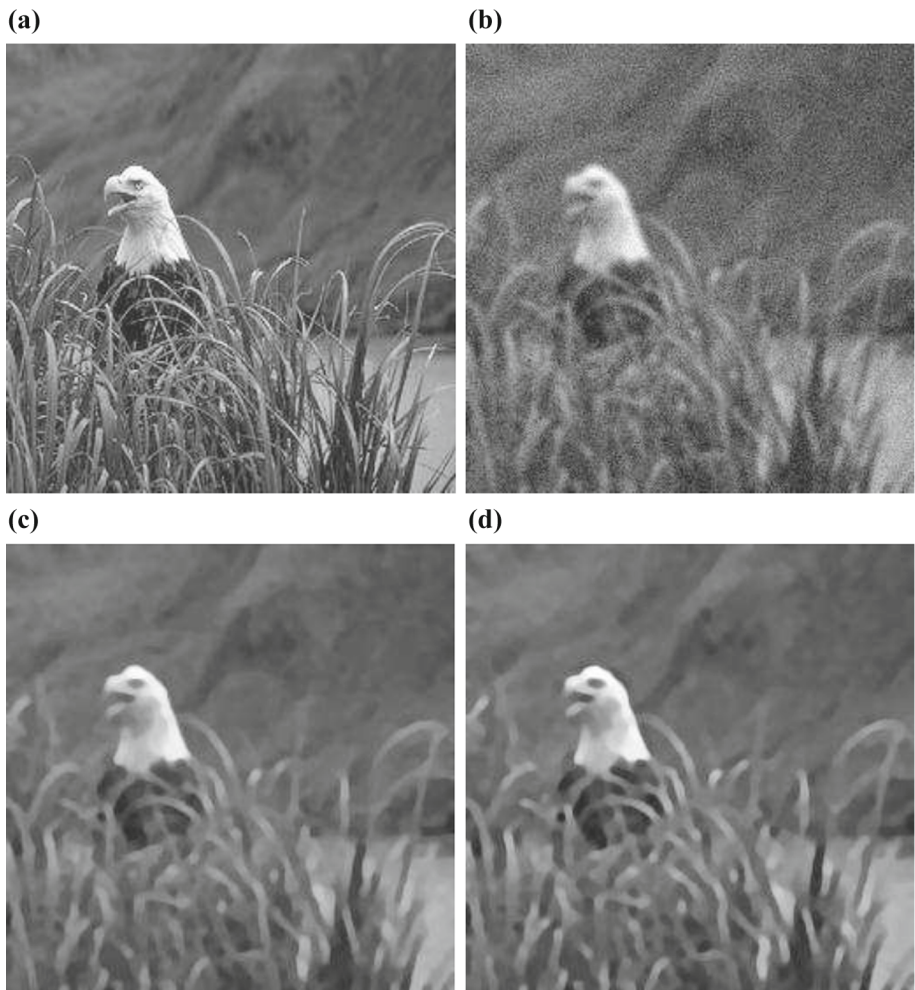


Fig. 10 Restoration results of comparison of TV model and our proposed model on Bird. **a** Original image; **b** damaged image with Gaussian kernel blur and Gaussian Noise of $\sigma = 10$ (PSNR=21.7525, CISI=0.4064); **c** TV model (PSNR=22.7110, CISI=0.5011); **d** our proposed model (PSNR=22.7482, CISI=0.5212)

fractional variable order differential model is more suitable for image processing than the previous image processing methods based on constant-order differential. In past published works, the same differential orders used to process different parts of an image. However, the differential orders are different and determined by the value of gradient modulus of the image in our proposed method. For example, when the pixel is located in smooth image area, the image gradient is very small and it causes the differential order α to be close to 1. This phenomenon is beneficial for denoising and suppressing the 'staircasing' effect. Conversely, when the pixel is located in image edges, the image gradient is large, and the differential order α should be a little larger value. This will preserve important texture. In addition, compared with our previous ones (Zhang et al. 2014b), in this paper, the original image is not only destroyed by white Gaussian noise, but also blurred by Gaussian kernel. Besides, in this

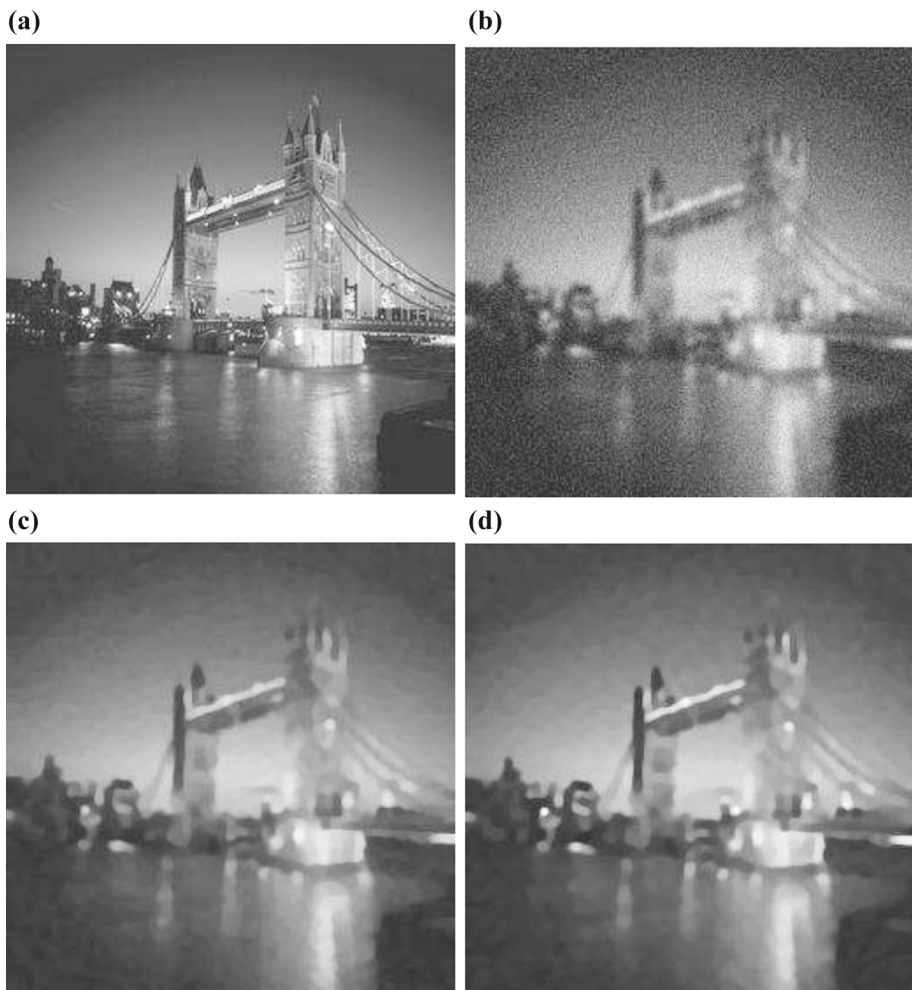


Fig. 11 Restoration results of comparison of TV model and our proposed model on Bridge. **a** Original image; **b** damaged image with Gaussian kernel blur and Gaussian Noise of $\sigma = 10$ (PSNR=23.4748, CISI=0.5198); **c** TV model (PSNR=25.3515, CISI=0.6560); **d** our proposed model (PSNR=25.4814, CISI=0.6757)

paper, our model increases the fidelity term $\frac{\lambda}{2} \int_{\Omega} (h * u - u_0)^2 d\Omega$. It can make the restored image retain the characteristics of the original image as much as possible.

Note that in the proposed Euler–Lagrange equation, when $\mathbf{A} = \mathbf{1}_E$ (a matrix of ones where every element is equal to one), Eq. (26) is equivalent to the corresponding Euler–Lagrange equation of total variation model in (You and Kaveh 2000).

4 The numerical implementation and simulation results

In this section, we illustrate the performance of our proposed method for image restoration. Combined image similarity index (CISI) and peak signal to noise ratio (PSNR) are used to

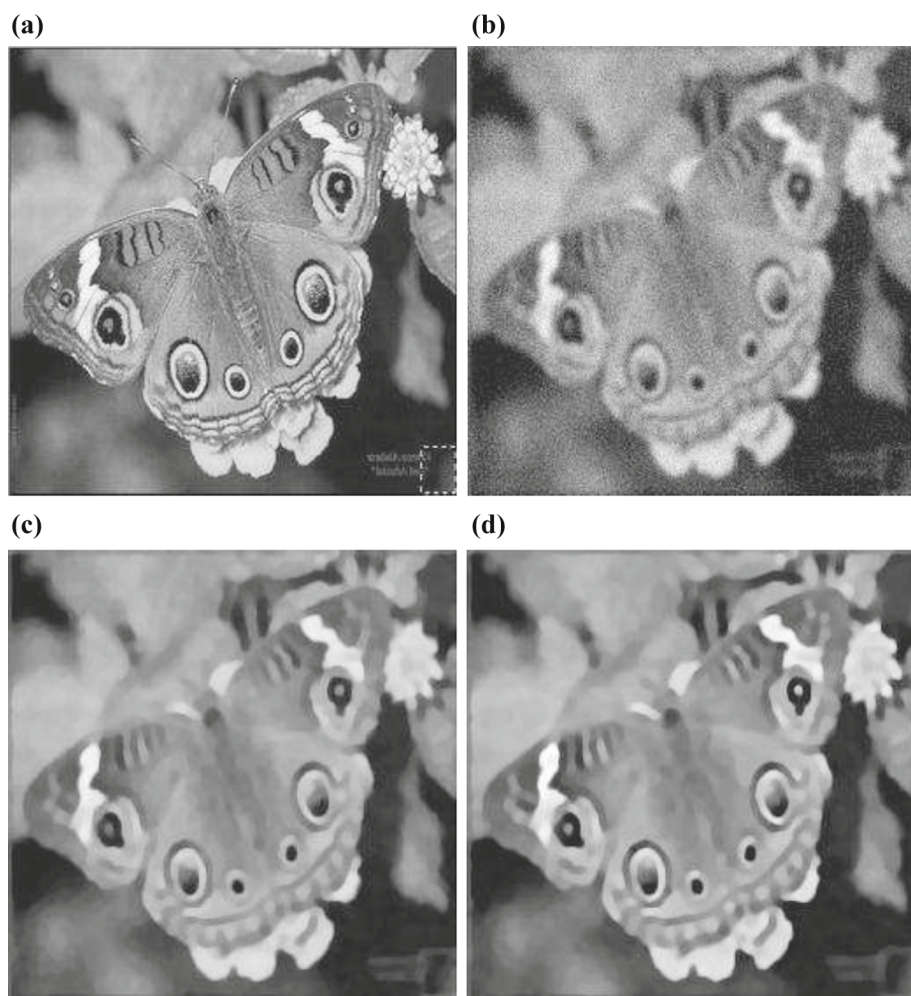


Fig. 12 Restoration results of comparison of TV model and our proposed model on Butterfly. **a** Original image; **b** damaged image with Gaussian kernel blur and Gaussian Noise of $\sigma = 10$ (PSNR=21.3074, CISI=0.3985); **c** TV model (PSNR=22.5801, CISI=0.5001); **d** our proposed model (PSNR=22.7271, CISI=0.5210)

measure the quality of the restored images (Okarma 2012, 2013; Zhang et al. 2011; Wang et al. 2004). The CISI is defined as follows:

$$CISI = (MS - SSIM)^a \cdot (VIF)^b \cdot (FSIM)^c,$$

where the MS-SSIM, VIF and FSIM denote Multi Scale Structural Similarity metric, Visual Information Fidelity metric and Feature Similarity metric, respectively. The exponent values ($a = 0.5$, $b = 0.3$, and $c = 5$) have also been obtained as the simplified result of the optimization procedure using the TID2008 database (chosen as the largest currently available dataset with many types of distortions) and further verified on the other databases.

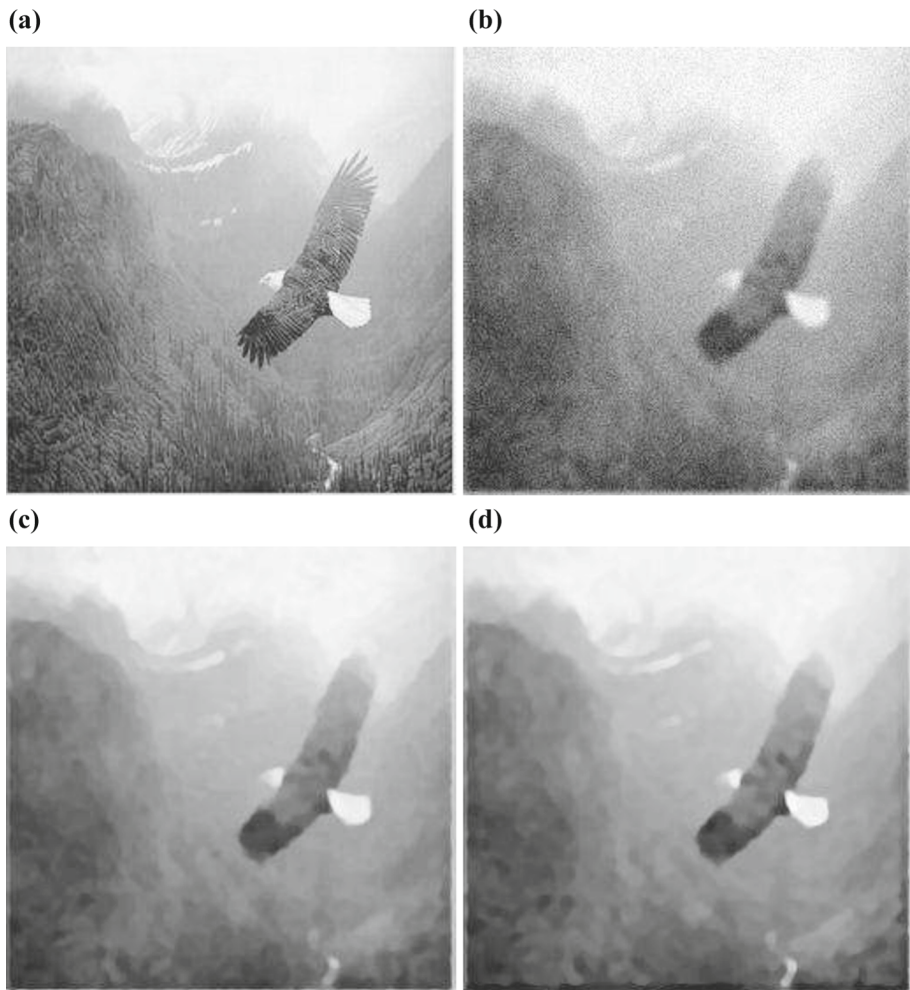


Fig. 13 Restoration results of comparison of TV model and our proposed model on Eagle. **a** Original image; **b** damaged image with Gaussian kernel blur and Gaussian Noise of $\sigma = 10$ (PSNR=23.7141, CISI=0.5201); **c** TV model (PSNR=26.0740, CISI=0.7031); **d** our proposed model (PSNR=26.7544, CISI=0.7253)

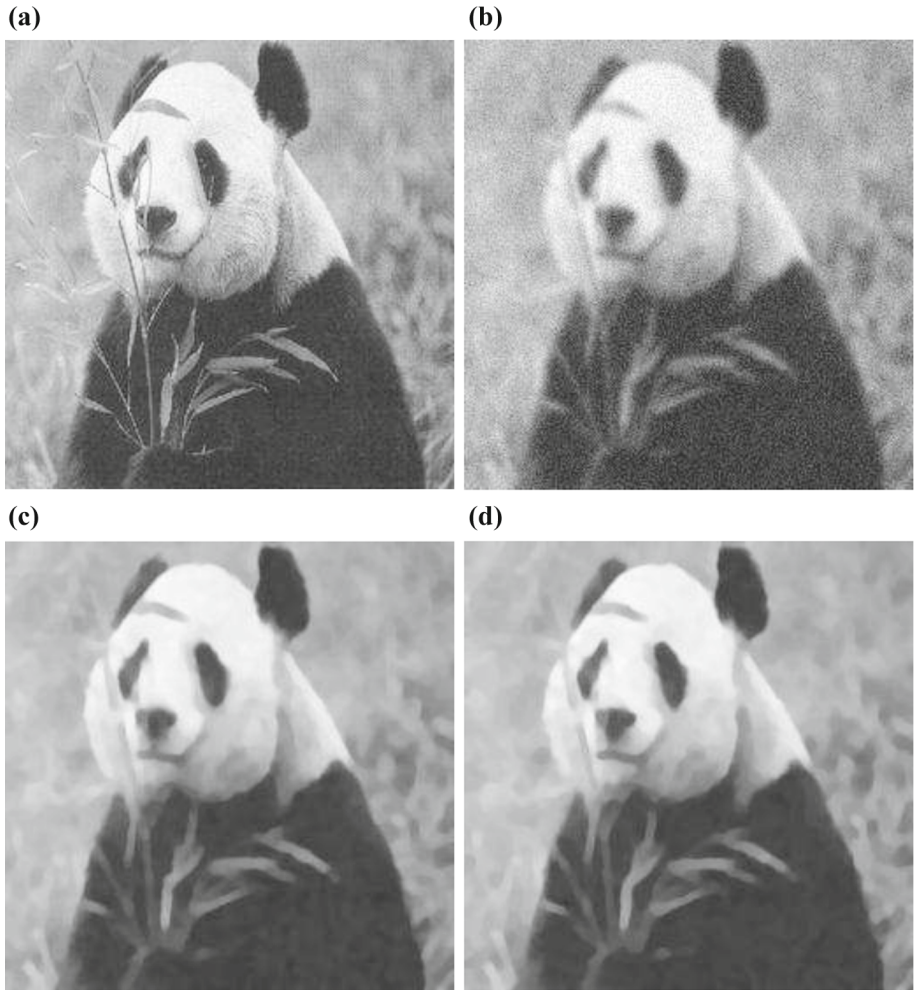


Fig. 14 Restoration results of comparison of TV model and our proposed model on Panda. **a** Original image; **b** damaged image with Gaussian kernel blur and Gaussian Noise of $\sigma = 10$ (PSNR=24.8799,CISI=0.5639); **c** TV model (PSNR=27.7442,CISI=0.7161); **d** our proposed model (PSNR=27.7705,CISI=0.7556)

The peak signal to noise ratio is defined as follows:

$$PSNR = 10 \times \log_{10} \left(\frac{255^2}{MSE} \right),$$

where $MSE = \frac{\|u - u_0\|_2^2}{M \times N}$ is the mean squared error, u_0 is the original image, u denotes the recovered image. The larger of PSNR is, the better the performance is.

To summarize, our image restoration approach is realized by the following steps.

- (1) Let the input image be u and set $n = 1$, $u_n = u$, $k, \Delta t$, $t = k \Delta t$, compute the 2-D DFT $\hat{u}_n$ of u_n .
- (2) Compute $|\nabla u_n|$, $\mathbf{A} = 2 \cdot \frac{|\nabla u_n| + 1}{|\nabla u_n| + 2}$, $\lambda = 2 - \mathbf{A}$.

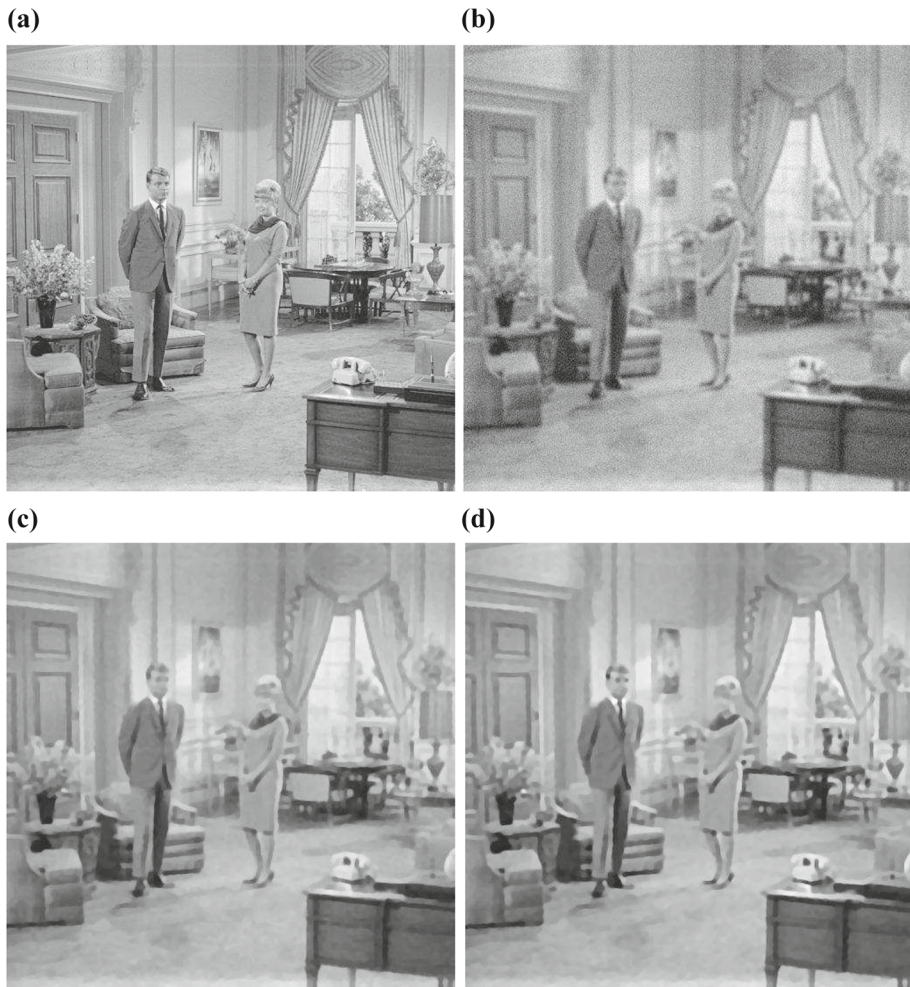


Fig. 15 Restoration results of comparison of TV model and our proposed model on Image43. **a** Original image; **b** damaged image with Gaussian kernel blur and Gaussian Noise of $\sigma = 10$ (PSNR=24.0936,CISI=0.5392); **c** TV model (PSNR=26.3094,CISI=0.7029); **d** our proposed model (PSNR=26.3640,CISI=0.7132)

(3) Compute $\mathbf{A}$ -order partial differences $\tilde{D}_{\mathbf{A}x}u_n$ and $\tilde{D}_{\mathbf{A}y}u_n$ by the formula:

$$\tilde{D}_{\mathbf{A}x}u_n = F^{-1}[(1 - \exp(-j2\pi\omega_1/m))^{\mathbf{A}} \exp(j\pi\mathbf{A}\omega_1/m)F(u_n)]$$

and

$$\tilde{D}_{\mathbf{A}y}u_n = F^{-1}[(1 - \exp(-j2\pi\omega_2/m))^{\mathbf{A}} \exp(j\pi\mathbf{A}\omega_2/m)F(u_n)],$$

compute

$$K_1^* = \text{diag}(\text{conj}((1 - \exp(-j2\pi\omega_1/m))^{\mathbf{A}} \exp(j\pi\mathbf{A}\omega_1/m))),$$

and

$$K_2^* = \text{diag}(\text{conj}((1 - \exp(-j2\pi\omega_2/m))^{\mathbf{A}} \exp(j\pi\mathbf{A}\omega_2/m))).$$

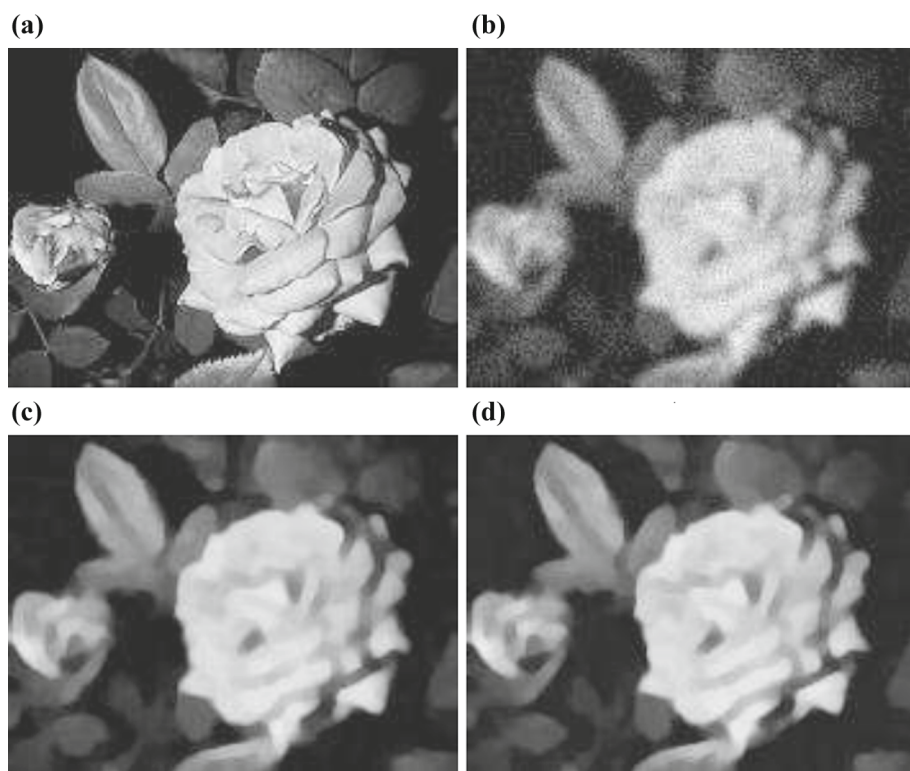


Fig. 16 Restoration results of comparison of TV model and our proposed model on Rose. **a** Original image; **b** damaged image with Gaussian kernel blur and Gaussian Noise of $\sigma = 10$ (PSNR=21.6974, CISI=0.4021); **c** TV model (PSNR=23.4043, CISI=0.5300); **d** our proposed model (PSNR=23.5005, CISI=0.5773)

Table 1 The comparison between the TV method and Our method

Images	Blurnoisly		TV method		Our method	
	PSNR	CISI	PSNR	CISI	PSNR	CISI
Changeone	24.1429	0.5211	25.9132	0.6770	26.2389	0.7032
Lena	24.1866	0.5301	27.0604	0.6932	27.3116	0.7161
Peppers	24.2929	0.5331	27.6985	0.7163	28.1243	0.7771
Lady	25.6635	0.6760	30.4625	0.8398	30.6072	0.8582
Fighter	27.2985	0.7407	35.1076	0.9359	35.4402	0.9467
Autumn	24.2184	0.5321	26.0972	0.7001	26.1313	0.7122
Bird	21.7525	0.4064	22.7110	0.5011	22.7482	0.5212
Bridge	23.4748	0.5198	25.3515	0.6560	25.4814	0.6757
Butterfly	21.3074	0.3985	22.5801	0.5001	22.7271	0.5210
Eagle	23.7141	0.5201	26.0740	0.7031	26.7544	0.7253
Panda	24.8799	0.5639	27.7442	0.7161	27.7705	0.7556
Image43	24.0936	0.5392	26.3094	0.7029	26.3640	0.7132
Rose	21.6974	0.4021	23.4043	0.5300	23.5005	0.5773

Table 2 Comparisons of BM3D, IRTV and Our method applied to test images corrupted by Gaussian kernel blur and Gaussian noise

Images	PSNR in dB		
	BM3D method	IRTV/p method	Our method
Carving	30.8978	30.9899/0.2	31.1199
Building	28.1121	28.2715/0.2	28.7988
Church	29.3129	29.0531/0.4	30.1171
Phantomdisk	33.0535	33.3260/0.2	33.6925
Satellite	28.0385	28.3856/0.6	28.5576
Nebula	23.4295	23.4681/0.2	24.5361

(4) Compute $h_{xn} = c(|\tilde{D}_A u_n|^2) \tilde{D}_A x u_n$ and $h_{yn} = c(|\tilde{D}_A u_n|^2) \tilde{D}_A y u_n$, then compute $\hat{g}_n = K_1^* \circ F(h_{xn}) + K_2^* \circ F(h_{yn})$, $\hat{T}_n = F[\lambda h * (h * F^{-1}(\hat{u}_n) - u_1)]$.

(5) Compute $\hat{u}_{n+1} = \hat{u}_n - (\hat{g}_n + \hat{T}_n) \cdot \Delta t$, and set $n = n + 1$; if $n = k$, compute the 2-D IDFT of $\hat{u}_n$, stop; else go to (2).

In our experiments, we take $\Delta t = 0.01$ and $k = 9$, and use the following function (Romeny 1994)

$$c(s) = \frac{1}{1 + (s/b_1)^2}$$

with $b_1 = 10$ in our experiment results.

4.1 Experiment 1: denoising and deblurring with fractional variable order differential

In this part of experimental studies, effectiveness of the fractional variable order differential method on denoising and deblurring performance was examined. We compare the proposed method with the TV image restoration method (Marquina and Osher 2000). The test 256×256 , 512×512 and 175×132 grayscale images include Changeone, Lena, Peppers, Lady, Fighter, Autumn, Bird, Bridge, Butterfly, Eagle, Panda, Image43 and Rose.

In Figs. 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 and 16, we separately list restored Changeone, Lena, Peppers, Lady, Fighter, Autumn, Bird, Bridge, Butterfly, Eagle, Panda, Image43 and Rose images obtained by our proposed model and the TV model. Figures 4a, 5a, 6a, 7a, 8a, 9a, 10a, 11a, 12a, 13a, 14a, 15a and 16a are the original images of the Changeone, Lena, Peppers, Lady, Fighter, Autumn, Bird, Bridge, Butterfly, Eagle, Panda, Image43 and Rose; Figs. 4b, 5b, 6b, 7b, 8b, 9b, 10b, 11b, 12b, 13b, 14b, 15b and 16b are damaged images of the Changeone, Lena, Peppers, Lady, Fighter, Autumn, Bird, Bridge, Butterfly, Eagle, Panda, Image43 and Rose with Gaussian kernel blur and Gaussian noise of $\sigma = 10$, and the corresponding PSNR and CISI are shown as Table 1. Figures 4c, 5c, 6c, 7c, 8c, 9c, 10c, 11c, 12c, 13c, 14c, 15c and 16c are the results by using modified TV method and the obtained PSNR and CISI are shown as Table 1. While using our method, the corresponding PSNR and CISI reach relatively high values shown as Table 1 [see Figs. 4d, 5d, 6d, 7d, 8d, 9d, 10d, 11d, 12d, 13d, 14d, 15d and 16d]. From the Table 1, we can confirm that compared our method with the TV method, the quality of the restored images becomes better. According to Figs. 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15 and 16 and Table 1, we can find that our proposed method is better than the TV method with respect to visual effect, PSNR and CISI.

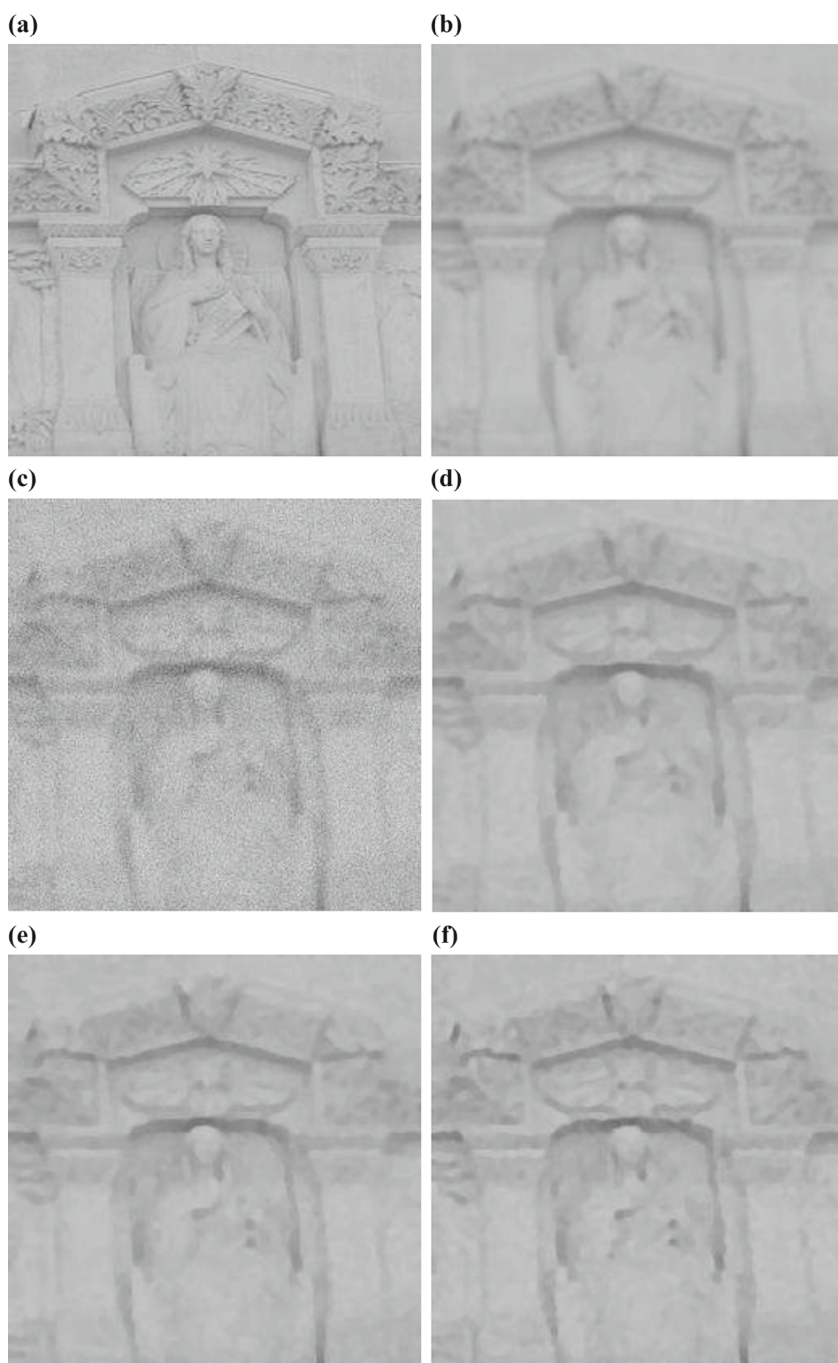


Fig. 17 Restoration results of comparison of BM3D method, IRTV method and our proposed method on Carving. **a** Original image; **b** damaged image with Gaussian kernel blur; **c** damaged image with Gaussian kernel blur and Gaussian Noise of $\sigma = 10$; **d** processed by BM3D method; **e** processed by IRTV method; **f** processed by our proposed method

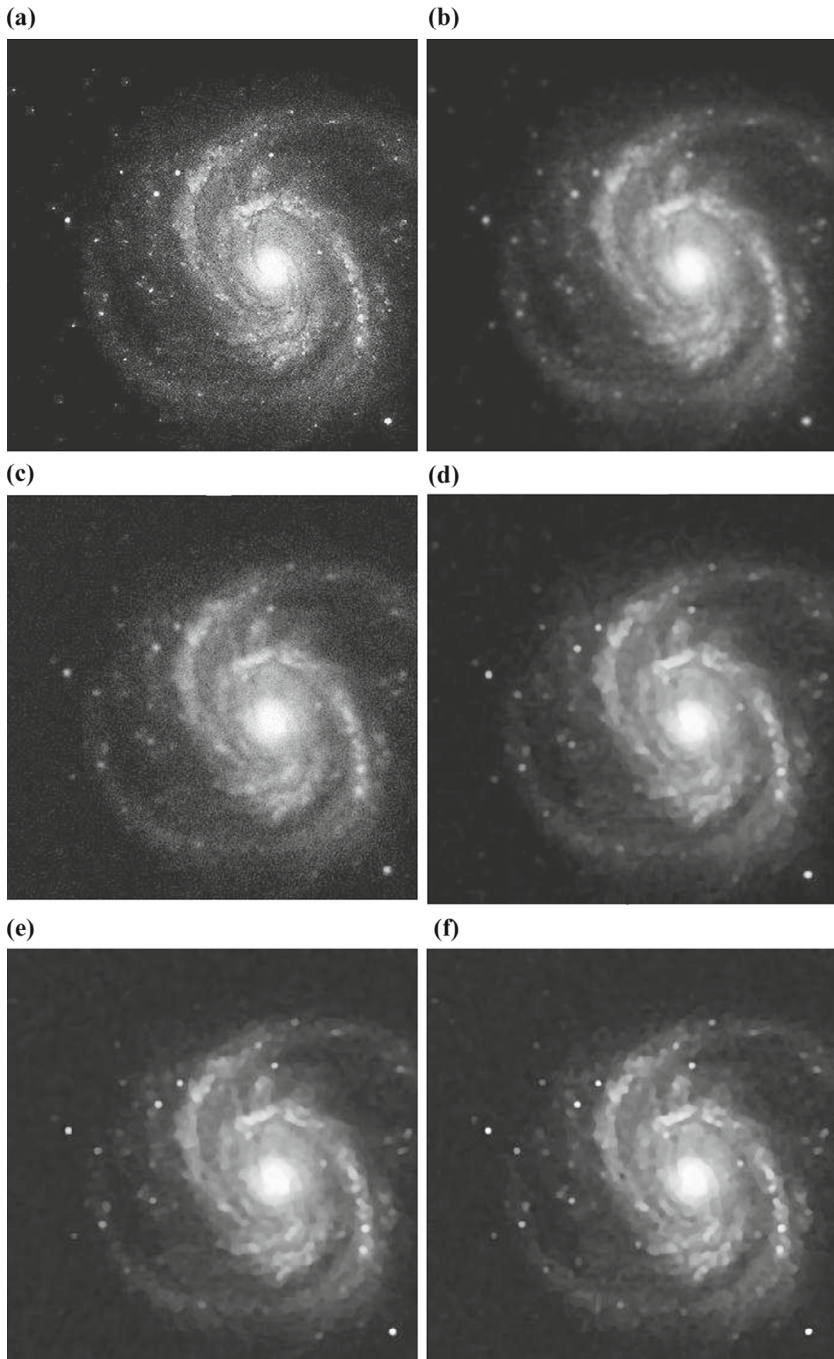


Fig. 18 Restoration results of comparison of BM3D method, IRTV method and our proposed method on Nebula. **a** Original image; **b** damaged image with Gaussian kernel blur; **c** damaged image with Gaussian kernel blur and Gaussian Noise of $\sigma = 10$; **d** processed by BM3D method; **e** processed by IRTV method; **f** processed by our proposed method

4.2 Experiment 2: comparisons of proposed method with IRTV and BM3D

In the literature, there are other known approaches to the image denoising. In Ref. (Yan and Lu 2015), an iteratively reweighted TV (IRTV) algorithm was proposed, and it was argued that the method offers several advantages for synthetic images in preserve edges as well as textures relative to TV model. Another novel image denoising strategy named BM3D based on enhanced sparse representation in transform domain was first proposed in Ref. (Dabov et al. 2007). The enhancement of the sparsity is achieved by grouping similar 2D image fragments into 3D data arrays, and a procedure of collaborative filtering is developed to deal with these 3D groups. As such, BM3D was found to offer state-of-the-art denoising performance. In what follows we report a set of experiments where the proposed fractional variable order differential model was carried out in comparison with the well-established IRTV and BM3D algorithms.

As in Experiment 1, the algorithms were applied to a variety of images of size 256×256 and 343×343 where each image was corrupted by Gaussian kernel and zero-mean Gaussian noise with standard deviation 10. By passing the standard deviation as a known parameter to the BM3D algorithm, we obtain the denoising results using MATLAB toolbox BM3D. The PSNR of the restored images obtained by these and fractional variable order differential method are given in Table 2. Here, we list typically the figures Carving and Nebula in Figs. 17 and 18.

From Table 2, it is observed that all three methods produce good results in PSNR, while our proposed method is better than other two methods with respect to visual effect and PSNR. This result is inevitable. Because using our proposed method (fractional variable order differential) to deal with images, the differential order can constantly change and adjust with different characteristics of images. It can not only remove noise, but also enhance texture and edge of images. Thus our method can also reducing the fuzzy for image.

5 Conclusions

In this paper, we have proposed a fractional variable order differential model for image restoration. The model can adaptively select the differential order according to the value of noise visibility of each pixel, which can effectively avoid staircase effect and the difficulties in parameter selection. In addition, this method is very easy to perform, and is a new idea of adaptive image processing. The experiment results show that this method is able to achieve a good effect in the respect of noise removal and edge preservation during the process of image smoothing. Future works involve extending the proposed method to other PDEs and variational models.

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